talked about being inspired by Ruby, Clojure, and Erlang. In fact, José was on the Rails Core Team, as well as the team that created the popular Devise library.

Elixir's language syntax is pleasant to use and elicits joy, much like Ruby's. For example, the Elixir core team works hard to avoid breaking changes—you can usually upgrade Elixir versions without too much work. That creates joy when upgrading from one version to another—an otherwise notoriously painful task.

Continual Innovation

Over the past few years, a ton of innovative features and libraries have been added to the Elixir ecosystem. These have been developed by the core team but also by private companies developing edge-pushing libraries.

The most well-known innovations in this space are Phoenix LiveView,[7] machine learning with Nx,[8] and hardware development with Nerves.[9]

A healthy foundation wouldn't matter if the technical aspects weren't also strong. Let's look at why Elixir is appealing for modern application development.

A Solid Foundation with the BEAM

Elixir is a functional programming language that's built on top of an almost forty-year-old foundation, the BEAM (Bogdan/Björn's Erlang Abstract Machine). We'll be going into what that is in the next section, but for now the important takeaway is that it's a unique runtime that empowers parallelism and fault tolerance in a way that isn't common in other languages.

The combination of fault tolerance and isolation is a premier strength of Elixir and the BEAM. Errors are going to happen, whether they are bugs, service outages, or anything in between. The BEAM provides ways for us to determine how errors should affect our application. Do you want an error to bring down the whole thing? Do you want an error to only affect the request that it was serving? Do you want to restart a piece of your system when an error occurs? Select the failure mode of your application so your application behaves predictably when things go wrong.

It's human nature to focus on the happy path. When evaluating a programming language, that would mean evaluating how you code in it, the libraries you use, and how data flows through your system. But the unhappy path is

7. https://github.com/phoenixframework/phoenix_live_view
8. https://github.com/elixir-nx
9. https://www.nerves-project.org/